10/12/2026: Smallest Counterexample

CSCI 246: Discrete Structures

Textbook reference: Sec 21, Scheinerman

Today's Agenda

- ($\approx$ 30 min) Mini-lecture
- ($\approx$ 20 min) Group exercises

The Well-Ordering Principle

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This set is a nonempty subset of the natural numbers. By the well-ordering principle, S contains a least element: ???

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- Conclude: The claim holds!

Proof by smallest counterexample: A data science example

Setting. Data often arrives as a **stream** (sensor readings, web logs, a file too big for memory). We can't store everything and then average, so we keep a **running mean**, updating it as each point x_n arrives:

$$\mu_1 = x_1, \qquad \mu_n = \mu_{n-1} + \frac{x_n - \mu_{n-1}}{n} \quad \text{for } n \geq 2.$$

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Why care? We've proven a one-pass streaming algorithm correct for *every* input length. ✓

The most famous proof by smallest counterexample

Prime Factorization Theorem (existence part of the Fundamental Theorem of Arithmetic)

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Why should we care? Internet Security! Your bank account is safe because every large number has a prime factorization, but nobody can find it.

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- What does that mean?

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- Then $s = ab$ is a product of primes, contradicting $s \in X$. ■

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Remark. This is the statement that "every positive integer has a binary representation." For example,

$$13 = 8 + 4 + 1 = 2^3 + 2^2 + 2^0 = 1101_{\text{two}}.$$

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Therefore X is empty, and the statement holds. ✓

Proof by the Well-Ordering Principle

Scheinerman's Proof Template 16.

Proof by the Well-Ordering Principle.

To prove a statement about natural numbers:

Proof. Suppose, for the sake of contradiction, that the statement is false. Let $X \subseteq \mathbb{N}$ be the set of counterexamples to the statement. (I like the letter X for eXceptions.) Since we have supposed the statement is false, $X \neq \emptyset$. By the Well-Ordering Principle, X contains a least element, x.

(Basis step.) We know that $x \neq 0$ because *show that the result holds for* 0*; this is usually easy.*

Consider $x - 1$. Since $x > 0$, we know that $x - 1 \in \mathbb{N}$ and the statement is true for $x - 1$ (because $x - 1 < x$). *From here we argue to a contradiction, often that x both is and is not a counterexample to the statement.*$\Rightarrow\Leftarrow$ ■

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Remarks.

- The basis step could be a different number than 0 (e.g. 1) if the statement is about a subset of the natural numbers (e.g. the positive integers).
- The contradiction needn't come from considering $x - 1$.

Group exercises

Ashley, Isabelle: 1
Bailey, Tanna: 5
Blaisdell, Olyver: 11
Brown, Ashton: 11
Brown, Jonathan: 6
Cardenas, Dominic: 12
Cutler, George: 2
Fabik, Roland: 9
Fellows, Audrey: 2
Fitzgerald, Ryan: 12
Fuller, Ryan: 9
Griffen, Clara: 4
Grutsch, Tanner: 11
Hager, Nathan: 6
Horner, Jayden: 3
Johnson, Jaret: 3
Kintzel, Samantha: 2
Lameres, Alexis: 13
Lee, Holden: 5
Mayala, Reece: 10
McLean, Connor: 7
Mcbride, Daniel: 10
Miller, Maverick: 8
Moss, Emily: 3
Mousseau, Matthew: 8
Neuman, Evan: 13
Putnam, John: 8
Ramos, Vydel: 7
Rathbun, Jedidiah: 1
Roberts, Matthew: 9
Roller, Nathan: 7
Ryan, Otis: 4
Shevtsov, Timothy: 6
Sinclair, Aidan: 5
Stensland, Emma: 12
Thiessen, Logan: 4
Thompson, Jaiden: 1
Vath, Ethan: 10

Group exercises

1. Use proof by smallest counterexample to show that $n < 2^n$ for all $n \in \mathbb{N}$.
2. Use proof by smallest counterexample to show that

$$1 + 2 + 3 + \cdots + n = \frac{1}{2}n^2 + \frac{1}{2}n$$

for all positive integers n.

Solution to group exercise #1

Problem. Use proof by smallest counterexample to show that $n < 2^n$ for all $n \in \mathbb{N}$.

Solution. Suppose, for the sake of contradiction, that the statement is false. Let $X \subseteq \mathbb{N}$ be the set of counterexamples to the statement. Since we have supposed the statement is false, $X \neq \emptyset$. So by the Well Ordering Principle, X contains a smallest element, s.

(Basis step). We know that $s > 0$ since $0 < 2^0 = 1$.

Since $s > 0$, the statement must be true for $s - 1$; that is:

$$s - 1 < 2^{s-1}$$

But then

$$\begin{aligned} s &< 2^{s-1} + 1 && \text{(add 1 to both sides)} \\ &\leq 2^{s-1} + 2^{s-1} && \text{(since } 1 \leq 2^{s-1}\text{, because } s - 1 \geq 0\text{)} \\ &= 2^s. \end{aligned}$$

So $s < 2^s$. So the statement is true for s. $\Rightarrow\Leftarrow$.

Solution to group exercise #2

Problem. Use proof by smallest counterexample to show that $1 + 2 + 3 + \cdots + n = \frac{1}{2}n^2 + \frac{1}{2}n$ for all positive integers n.

Solution. Suppose, for the sake of contradiction, that the statement is false. Let $X \subseteq \mathbb{N}$ be the set of counterexamples to the statement. Since we have supposed the statement is false, $X \neq \emptyset$. So by the Well Ordering Principle, X contains a smallest element, s.

(Basis step). We know that $s > 1$ since $1 = \frac{1}{2} \cdot 1^2 + \frac{1}{2} \cdot 1$.

Since $s > 1$, the statement must be true for $s - 1$; that is:

$$1 + 2 + 3 + \cdots + s - 1 = \frac{1}{2}(s-1)^2 + \frac{1}{2}(s-1)$$

So, adding s to the LHS and RHS, we have

$$\begin{aligned} 1 + 2 + 3 + \cdots + s - 1 + s &= \frac{1}{2}(s-1)^2 + \frac{1}{2}(s-1) + s \\ &= \frac{1}{2}(s^2 - 2s + 1) + \frac{1}{2}s - \frac{1}{2} + s \\ &= \frac{1}{2}s^2 + \cancel{-s} + \cancel{\frac{1}{2}} + \frac{1}{2}s + \cancel{-\frac{1}{2}} + \cancel{s} \\ &= \frac{1}{2}s^2 + \frac{1}{2}s \end{aligned}$$

So the statement is true for s. $\Rightarrow\Leftarrow$.

Appendix

Optional Remark About Group Exercise #2

The claim of group exercise # 2 can be presented slightly differently as:

$$1 + 2 + 3 + \cdots + n = \binom{n+1}{2} \text{ for all positive integers } n.$$

Note that

$$\binom{n+1}{2} = \frac{(n+1)!}{2!(n-1)!} = \frac{(n+1)n}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$$

So the alternative version is equivalent to the original version.